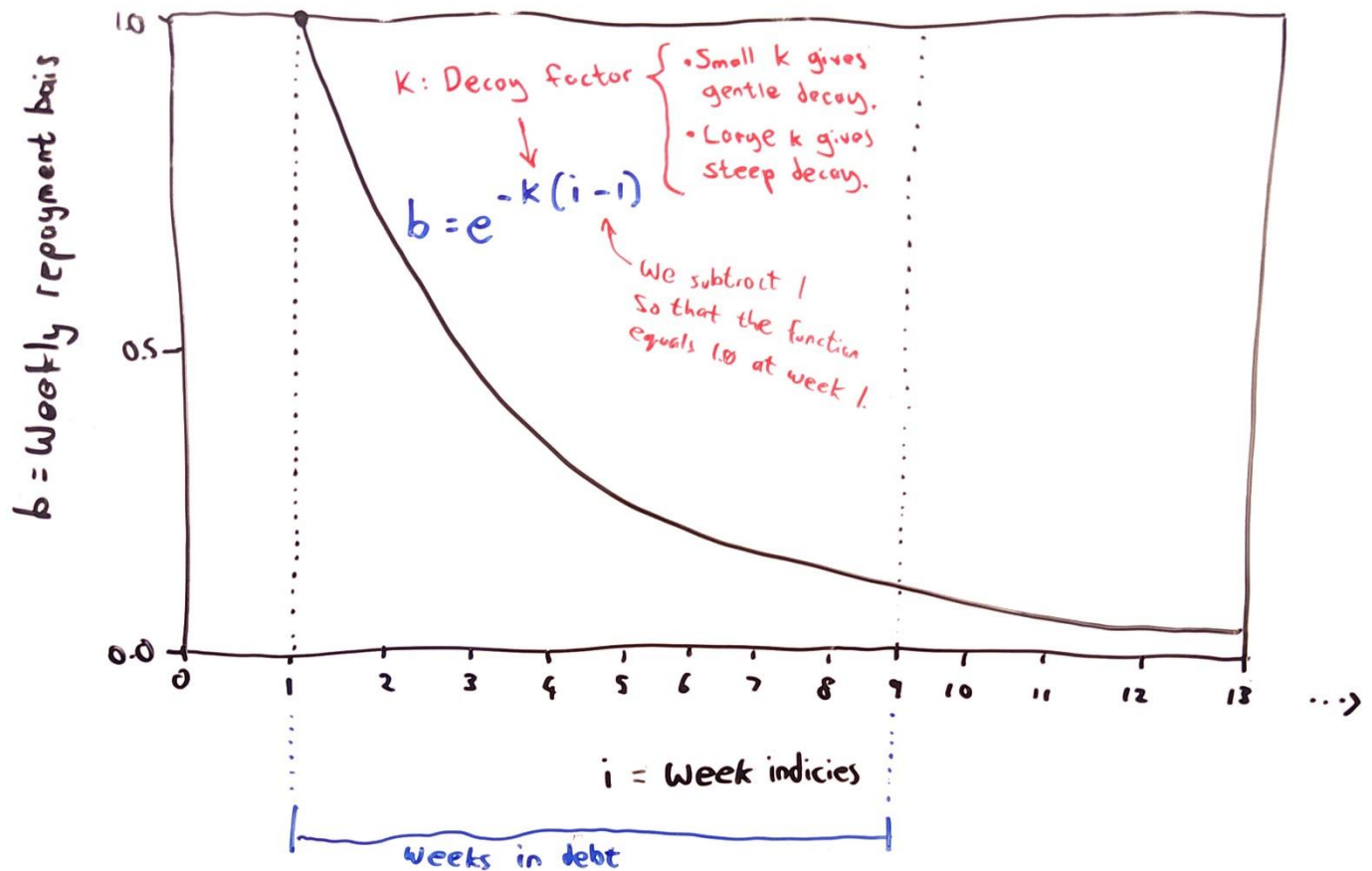


Exponentially Biased Debt Repayment Weight:

- Used to rank indebted users, based on historical repayment behavior.
- An exponential decay function may be used to emphasise recent repayment behavior, while still acknowledging past repayment behavior to a lesser degree.



Hint:

- Use [wolframalpha.com](https://www.wolframalpha.com) to quickly plot a graph of $b = e^{-k(i-1)}$
- Example:

WolframAlpha computable answers

plot($y = e^{(-1.0*(x-1))}$, $x=1$ to 10 , $b=0$ to 1)

This is k

- Use a value of 1.0 by default.
- Then try larger values, eg. 2.0, 3.5, 8.0, ... etc.
- Then try smaller values, eg. 0.5, 0.35, 0.01, ... etc

Note:

- We use x and y in Wolfram instead of i and b , because Wolfram will automatically interpret i as an imaginary number. Which we don't want.

Compute normalized percentage of outstanding debt repayed in week i :

L = Total debt to date.

P_i = Amount repayed in week i .

$$d_i = \frac{P_i}{L}$$

... Normalized percentage of outstanding debt repayed in week i .

• Multiply by 100 to get percentage.

• But, use the normalized value $\frac{P_i}{L}$ for computations.

Compute exponentially biased debt repayed in week i :

$$b_i = d_i \cdot e^{-k(i-1)}$$

... Where

• k is a manually chosen decay factor.

- A small value of k results in a gentle decay that increases emphasis of historical repayment behavior.

- A large value of k results in a steep decay that decreases emphasis of historical repayment behavior.

• i is the index of the week. ie. $i=1$ is week 1, $i=2$ is week 2, etc.

• d_i is the normalized percentage of outstanding debt repayed in week i .

• e = Euler's Number $\approx 2.71828...$

Compute user's debt weight:

$$w = \sum_{i=1}^n b_i$$

... Where

• n is the number of historical weeks to include in the sum.

• i is the index of the week. $i=1$ is last week, $i=2$ is the week before that, etc.

• b_i is the exponentially biased debt repayed in week i .

Updated version of the exponential decay function that allows it to return exactly zero at a specific week index.

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g file.

n = Number of weeks to include.

k = Decay factor

$$C_0 = \frac{e^k}{e^k - e^{kn}}$$

$$C_1 = \frac{e^{kn}}{e^{kn} - e^k}$$

$y = 1$ when $x = 1$

$y = 0$ when $x = n$

$$y = C_0 + C_1 \cdot e^{-k(x-1)}$$

These two factors are added to enable us to specify at which week the weight should be 0.

Example:

